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revolucionamos la comunicación

TECNOLOGÍAS DEL BIG DATA

EJEMPLOS ARQUITECTURAS DE DATOS



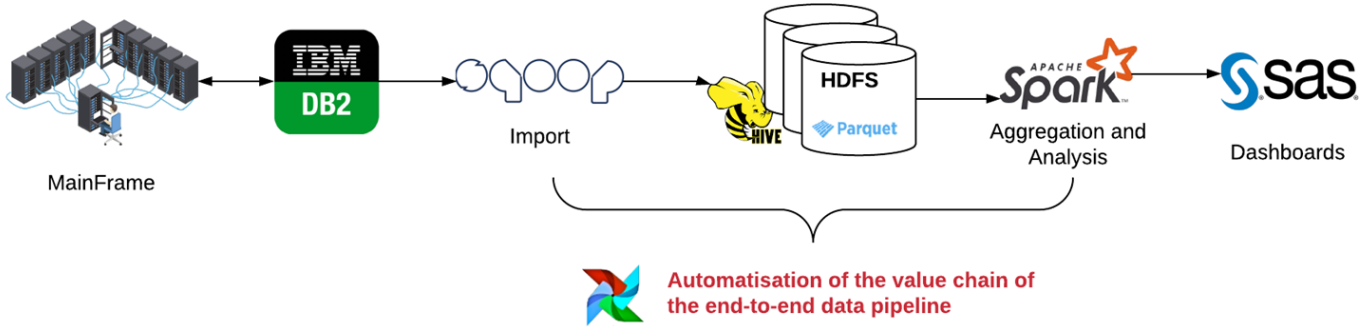
Agenda

- Ejemplos

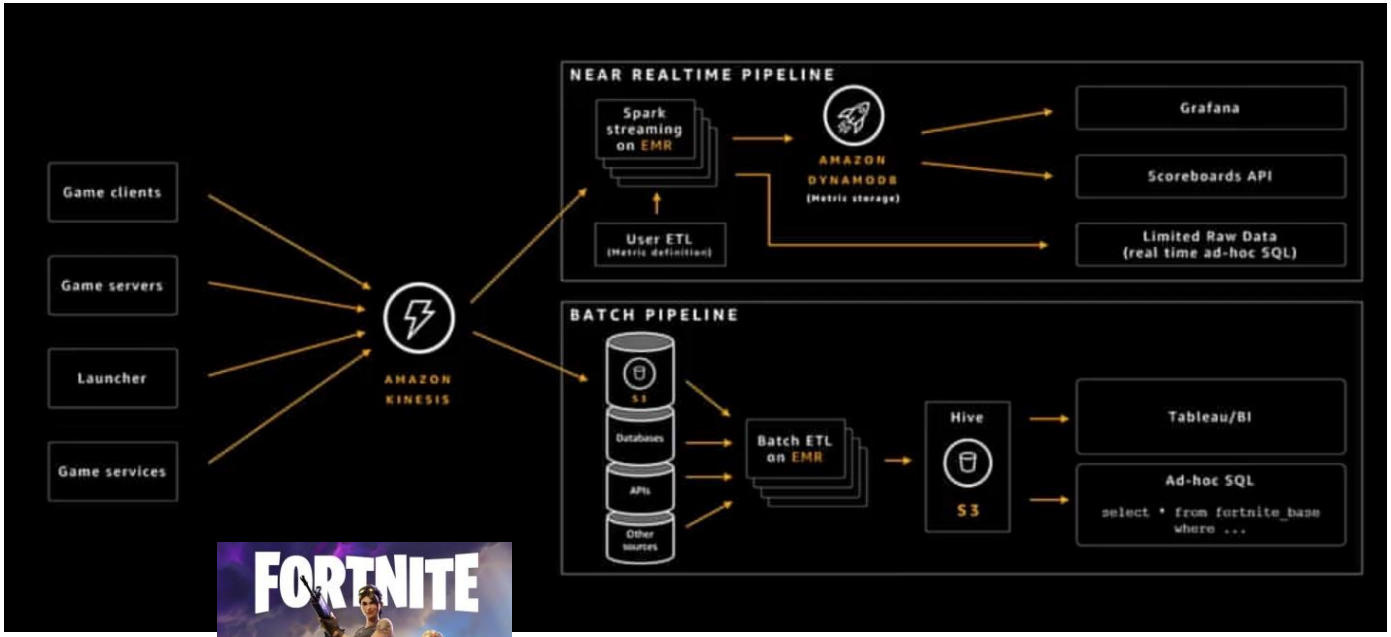
1.

EJEMPLOS

Arquitectura Batch (On Premise)

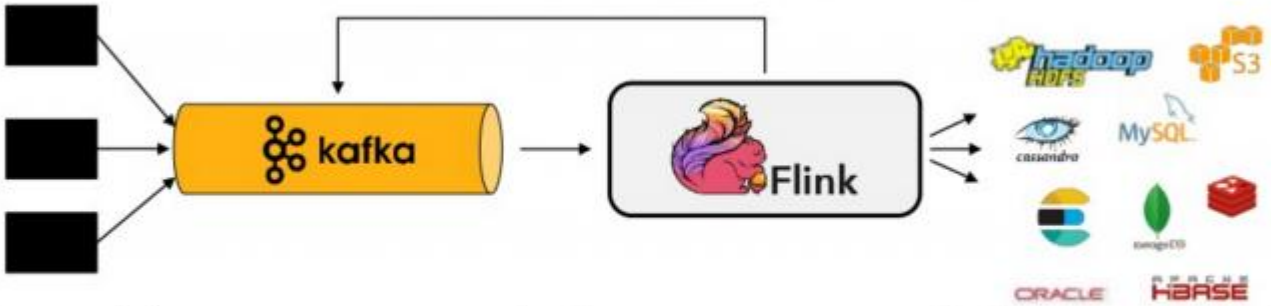


Arquitectura Lambda (AWS)





Arquitectura Kappa

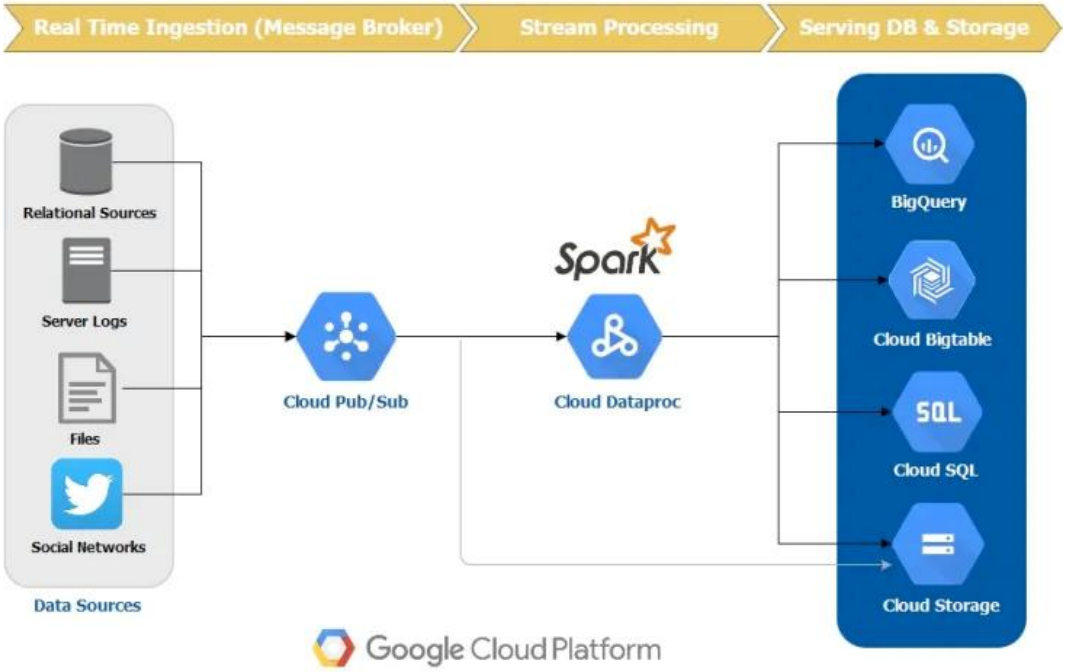


streaming
storage

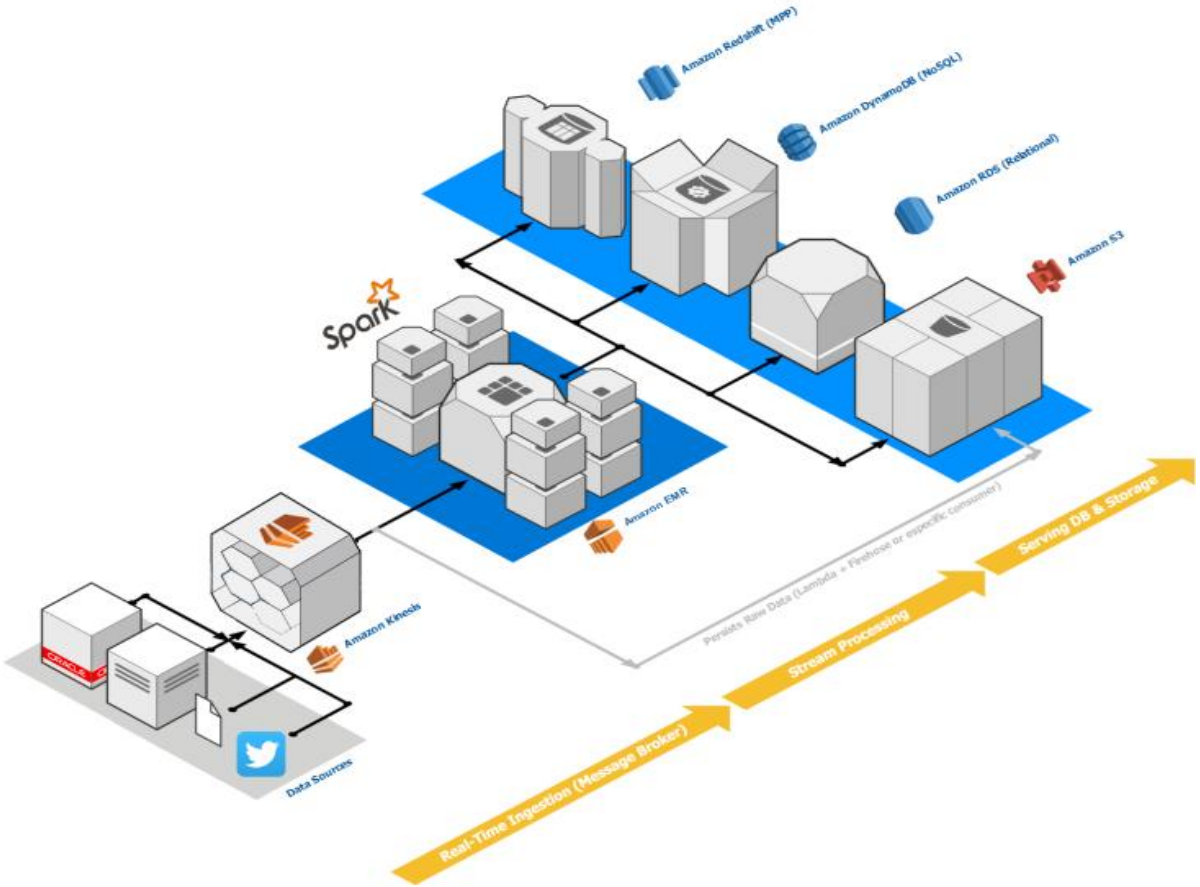
streaming
processing

serving

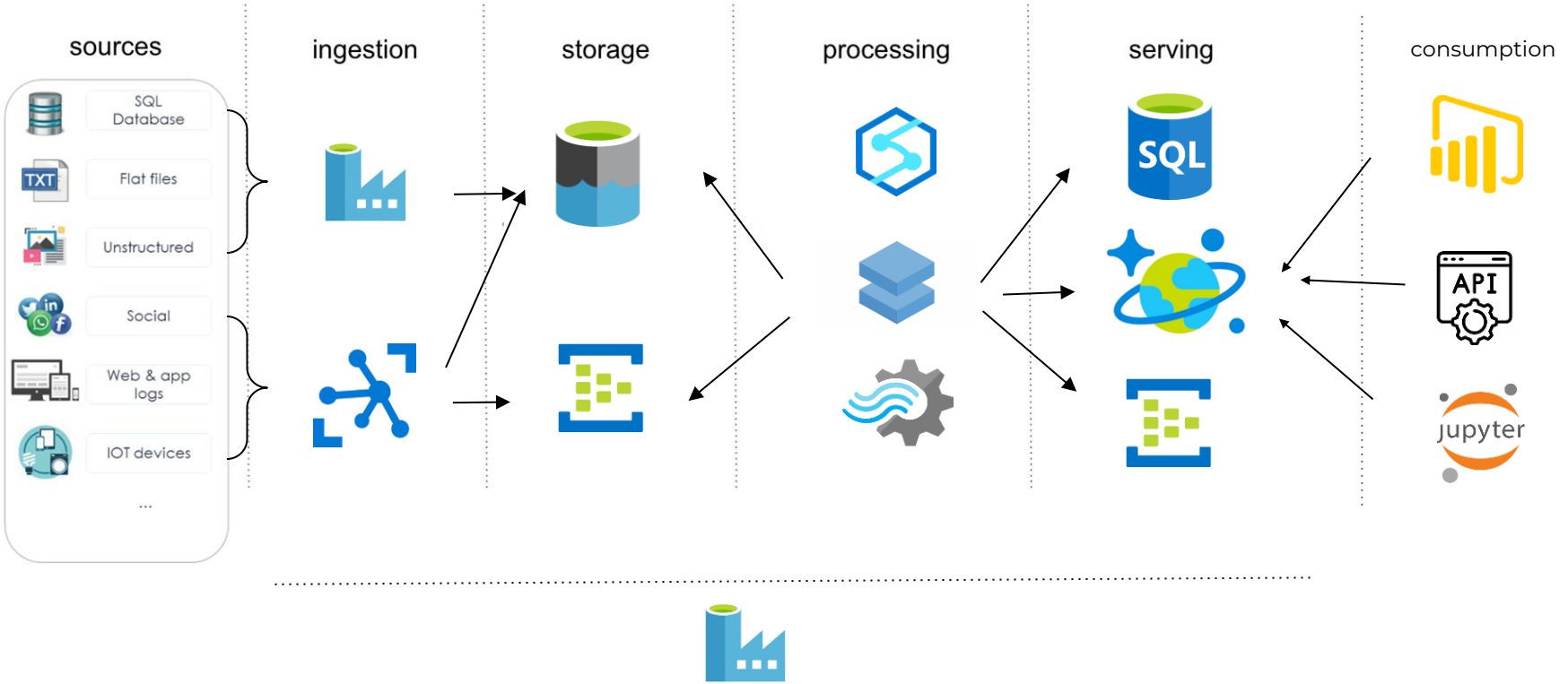
Arquitectura Kappa (GCP)



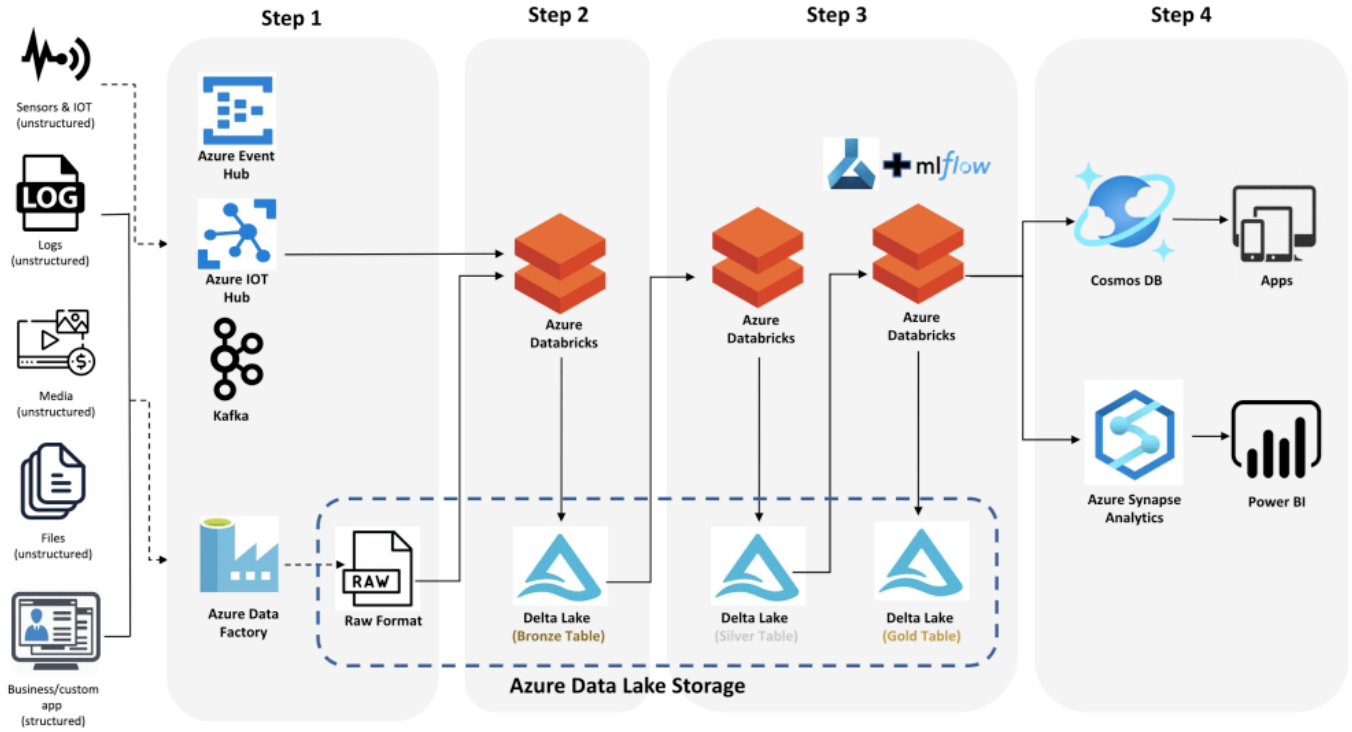
Arquitectura Kappa (AWS)



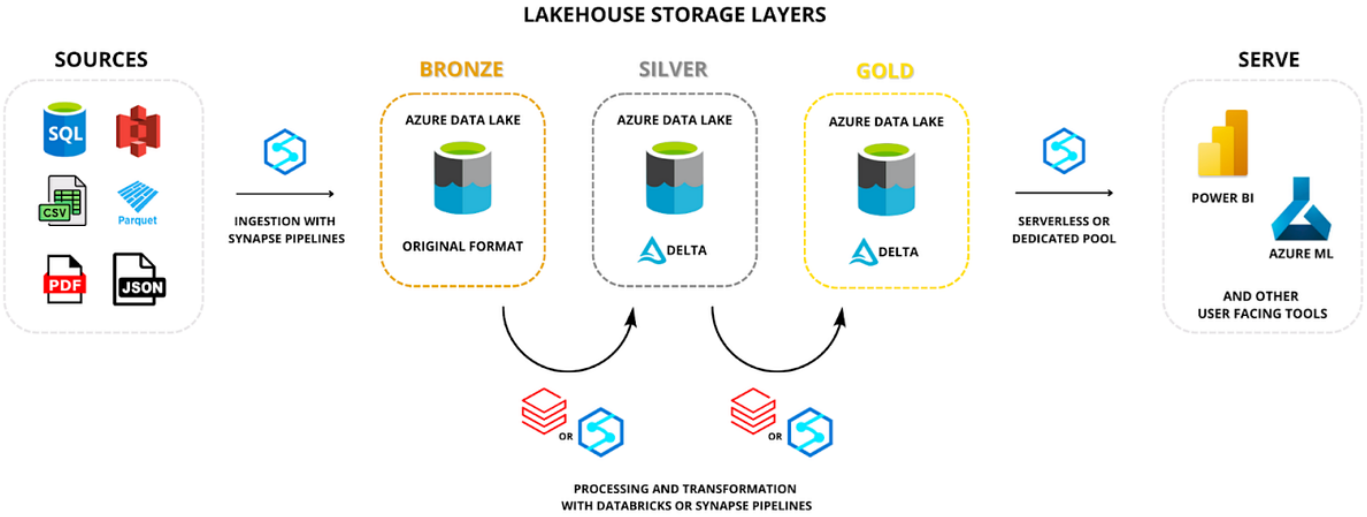
Arquitectura Azure



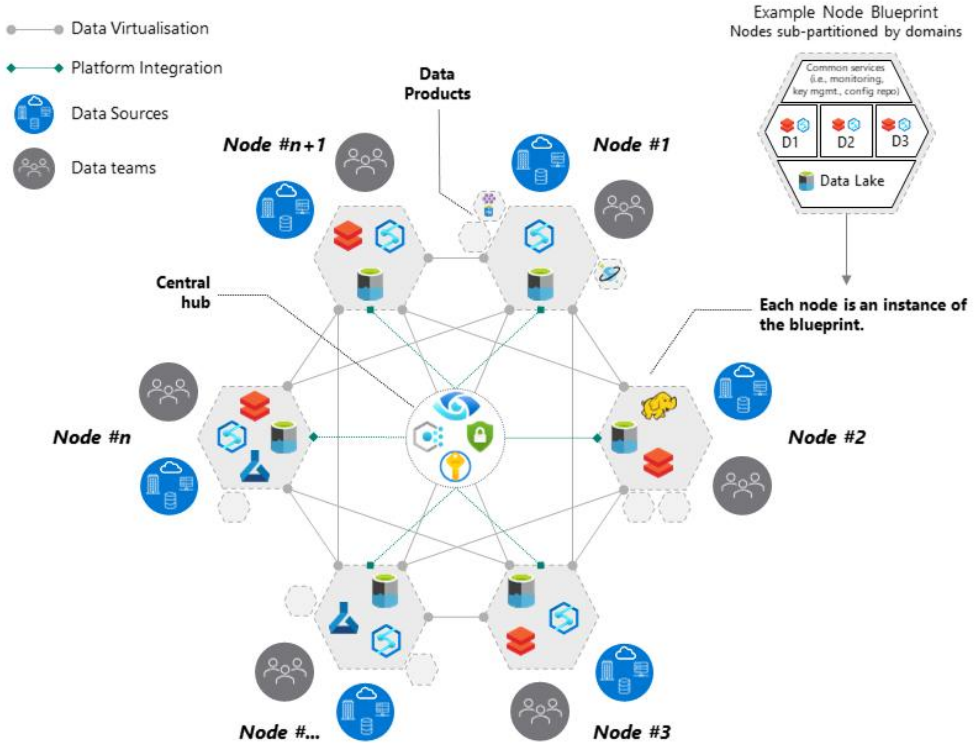
Arquitectura Delta (Azure)



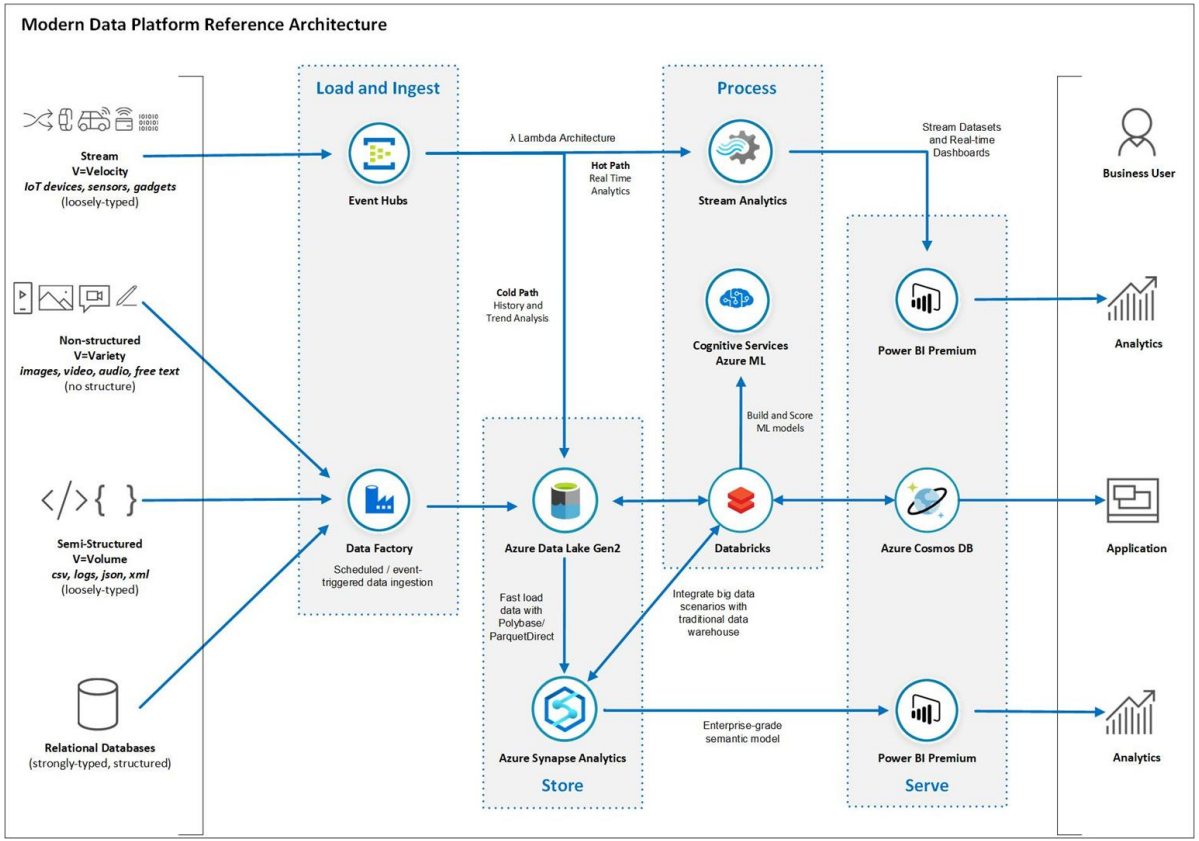
Arquitectura Delta (Azure)



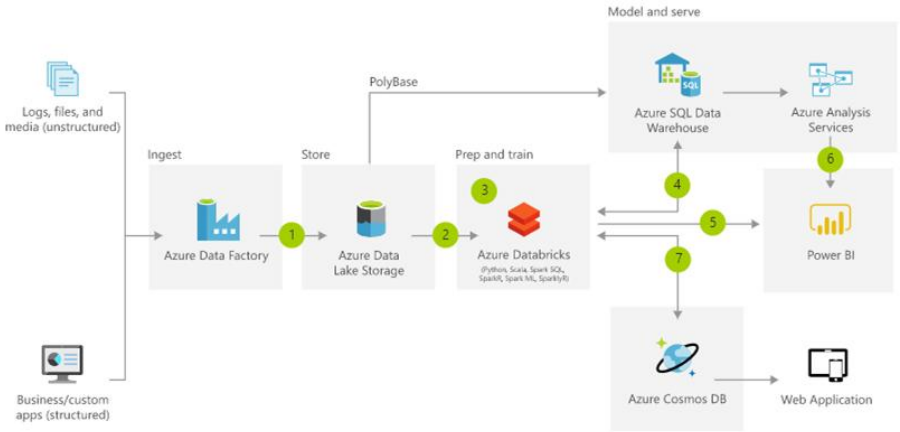
Arquitectura Data Mesh (Azure)



Arquitectura Datos (Azure)

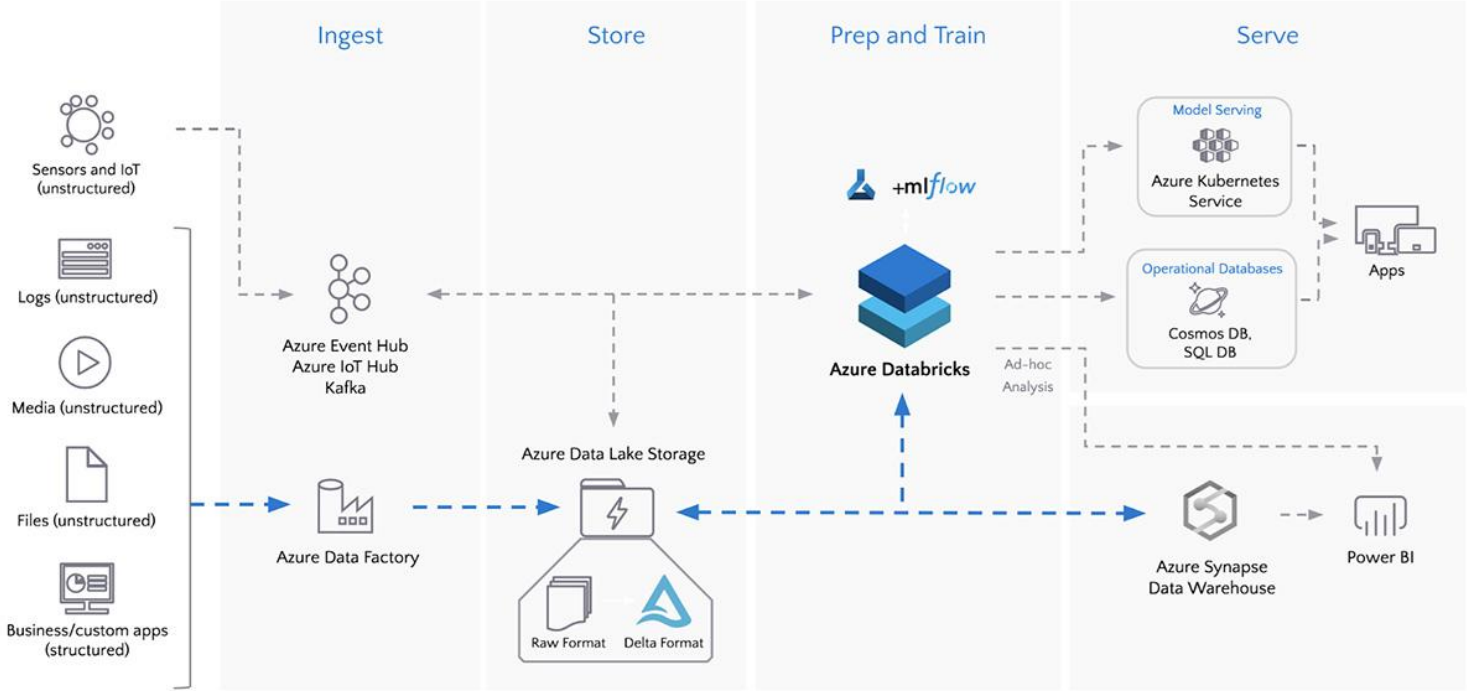


Arquitectura Analítica Avanzada (Azure)

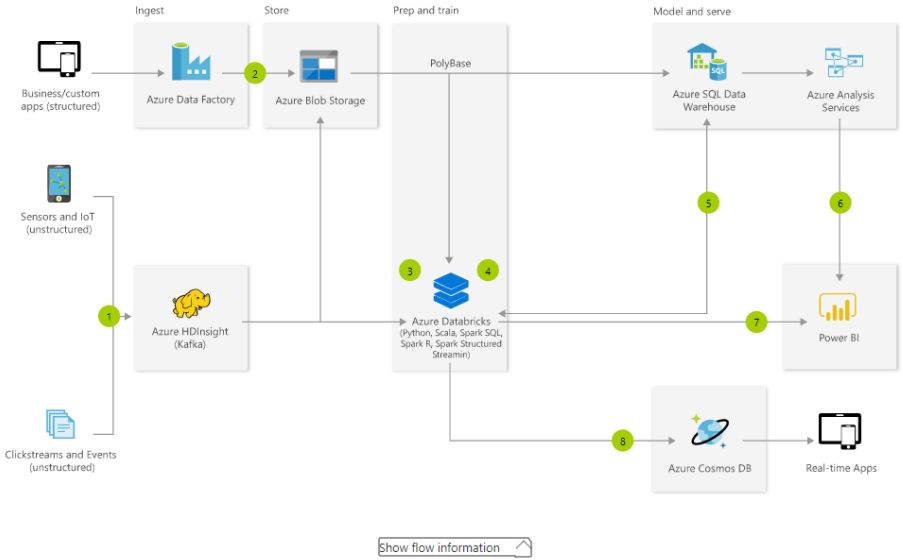


1. Bring together all your structured, unstructured and semi-structured data (logs, files and media) using Azure Data Factory to Azure Blob Storage.
2. Use Azure Databricks to clean and transform the structureless datasets and combine them with structured data from operational databases or data warehouses.
3. Use scalable machine learning/deep learning techniques, to derive deeper insights from this data using Python, R or Scala, with inbuilt notebook experiences in Azure Databricks.
4. Leverage native connectors between Azure Databricks and Azure SQL Data Warehouse to access and move data at scale.
5. Power users take advantage of the inbuilt capabilities of Azure Databricks to perform root cause determination and raw data analysis.
6. Run ad hoc queries directly on data within Azure Databricks.
7. Take the insights from Azure Databricks to Cosmos DB to make them accessible through web and mobile apps.

Arquitectura ML/AI (Azure)



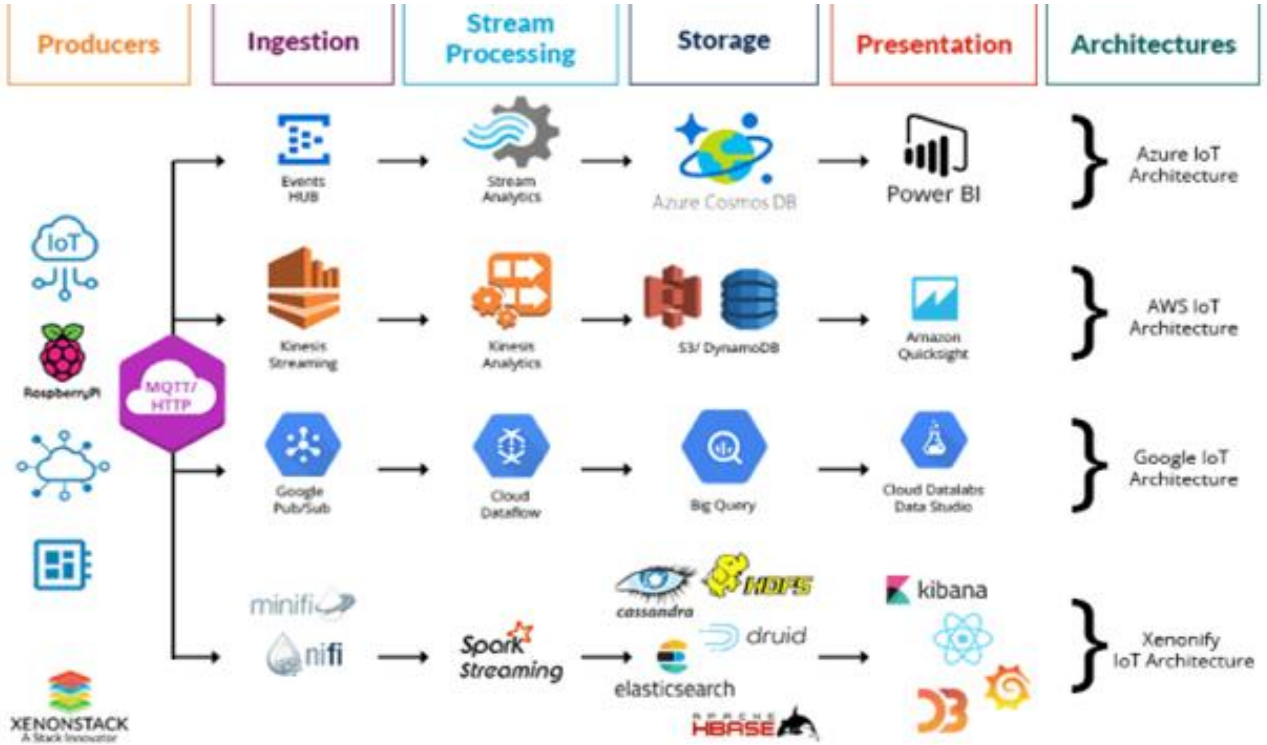
Arquitectura Tiempo Real (Azure)



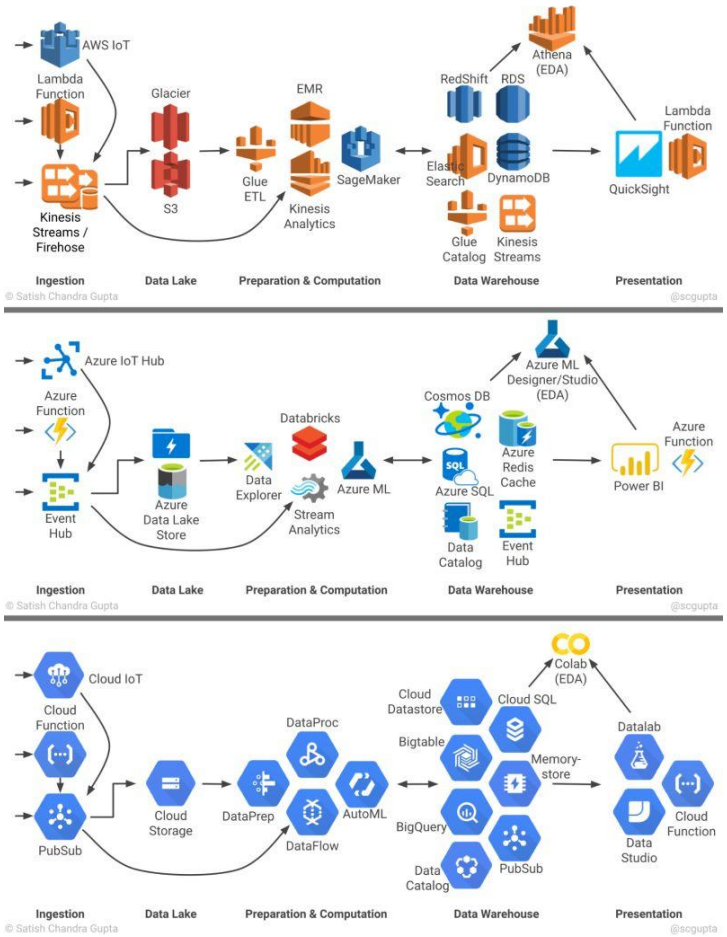
Show flow information

1. Easily ingest live streaming data for an application using Apache Kafka cluster in Azure HDInsight.
2. Bring together all your structured data using Azure Data Factory to Azure Blob Storage.
3. Take advantage of Azure Databricks to clean, transform and analyze the streaming data and combine it with structured data from operational databases or data warehouses.
4. Use scalable machine learning/deep learning techniques, to derive deeper insights from this data using Python, R or Scala, with inbuilt notebook experiences in Azure Databricks.
5. Leverage native connectors between Azure Databricks and Azure SQL Data Warehouse to access and move data at scale.
6. Build analytical dashboards and embedded reports on top of Azure Data Warehouse to share insights within your organization and use Azure Analysis Services to serve this data to thousands of users.
7. Power users take advantage of the inbuilt capabilities of Azure Databricks and Azure HDInsight to perform root cause determination and raw data analysis.
8. Take the insights from Azure Databricks to Cosmos DB to make them accessible through real time apps.

Arquitectura IoT



Big Data Pipelines on AWS, Microsoft Azure, and GCP



Data Lakehouse – Microsoft Fabric

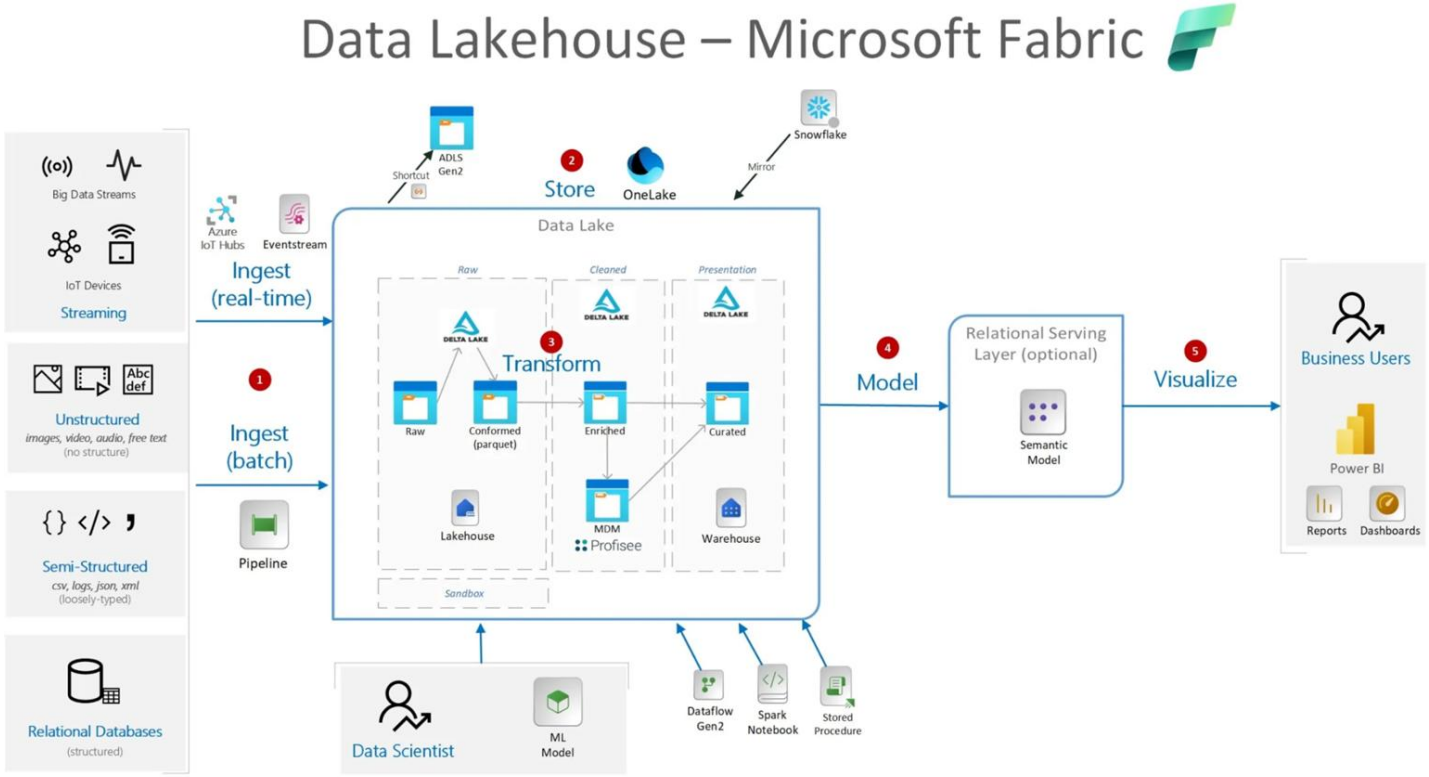
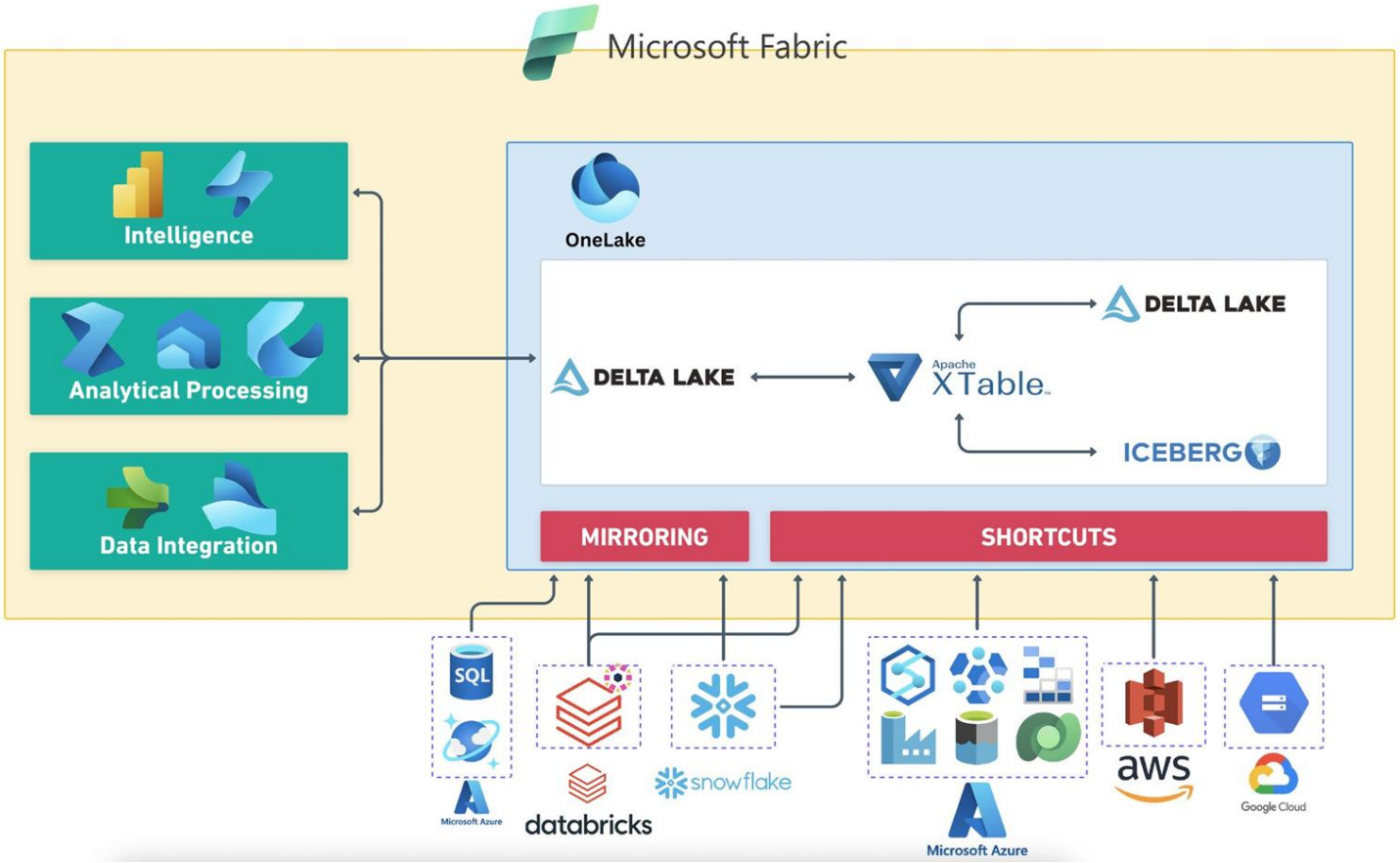
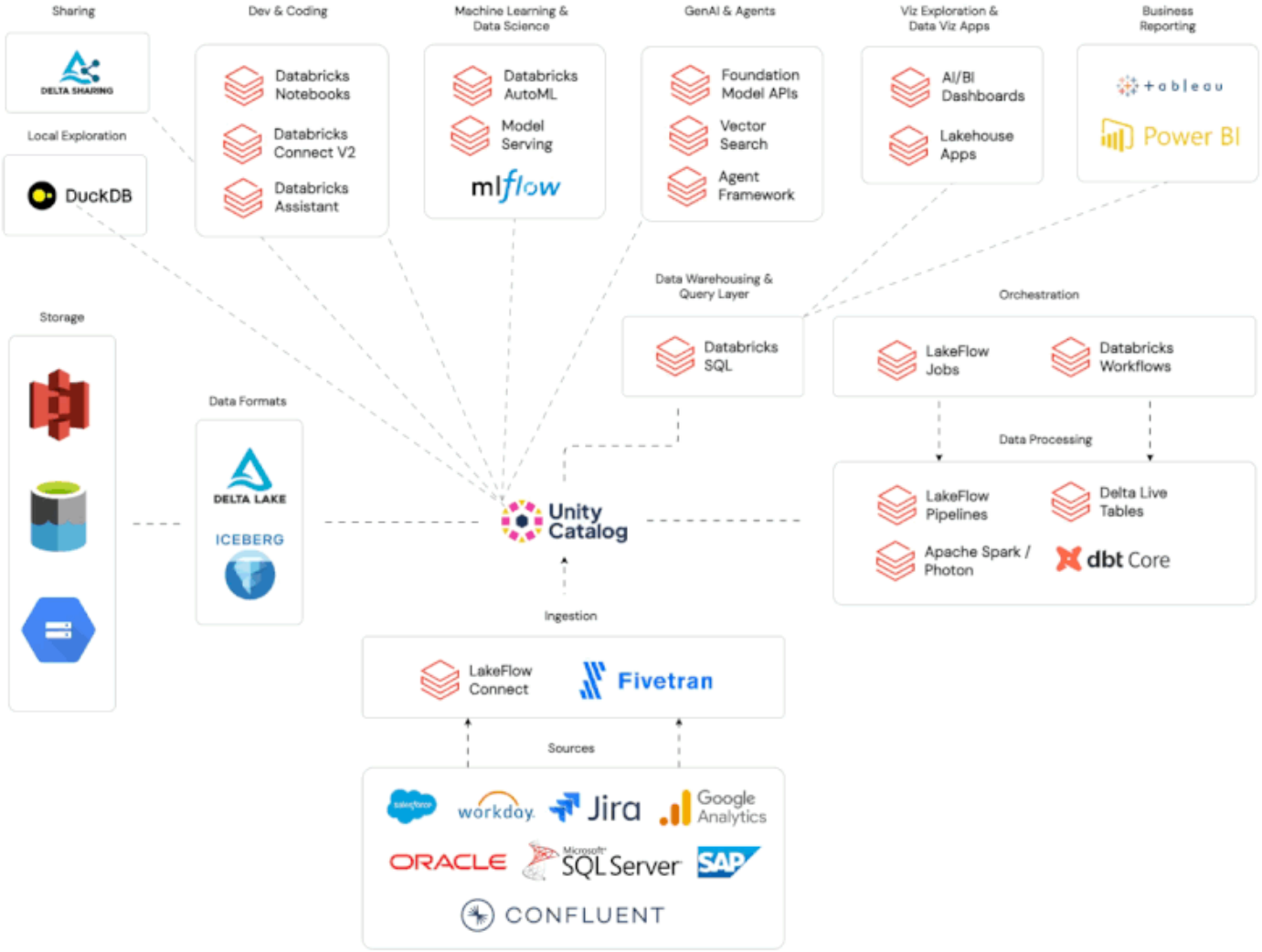


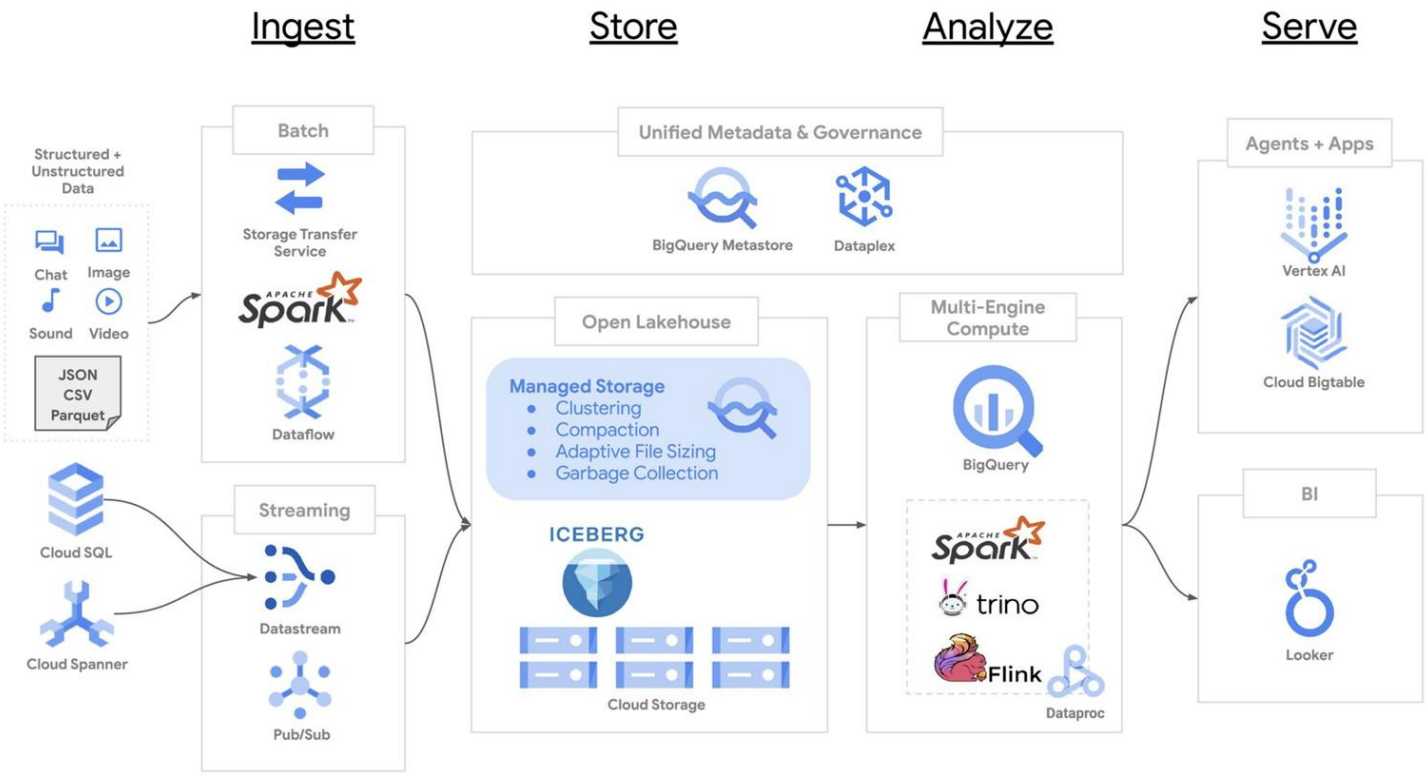
Diagram by ***JamesSerra.com***

Discover & Govern — Microsoft Purview

Platform — Microsoft Entra ID | Cost Management | Azure Key Vault | Azure Monitor | Security Center | Azure DevOps & GitHub | Azure Policy


Microsoft Fabric
Intelligence
Analytical Processing
Data Integration
OneLake
DELTA LAKE
DELTA LAKE
Apache XTable
ICEBERG
MIRRORING
SHORTCUTS
SQL
Microsoft Azure
databricks
snowflake
Microsoft Azure
aws
Google Cloud


Sharing
DELTA SHARING
Local Exploration
DuckDB
Dev & Coding
Databricks Notebooks
Databricks Connect V2
Databricks Assistant
Machine Learning & Data Science
Databricks AutoML
Model Serving
mlflow
GenAI & Agents
Foundation Model APIs
Vector Search
Agent Framework
Viz Exploration & Data Viz Apps
AI/BI Dashboards
Lakehouse Apps
Business Reporting
tableau
Power BI
Storage
Data Formats
DELTA LAKE
ICEBERG
Data Warehousing & Query Layer
Databricks SQL
Orchestration
LakeFlow Jobs
Databricks Workflows
Data Processing
LakeFlow Pipelines
Delta Live Tables
Apache Spark / Photon
dbt Core
Unity Catalog
Ingestion
LakeFlow Connect
Fivetran
Sources
salesforce
workday
Jira
Google Analytics
ORACLE
Microsoft SQL Server
SAP
CONFLUENT


Ingest
Store
Analyze
Serve
Structured + Unstructured Data
Chat
Image
Sound
Video
JSON
CSV
Parquet
Cloud SQL
Cloud Spanner
Batch
Storage Transfer Service
Spark
Dataflow
Streaming
Datastream
Pub/Sub
Unified Metadata & Governance
BigQuery Metastore
Dataplex
Open Lakehouse
Managed Storage
Clustering
Compaction
Adaptive File Sizing
Garbage Collection
ICEBERG
Cloud Storage
Multi-Engine Compute
BigQuery
Spark
trino
Flink
Dataproc
Agents + Apps
Vertex AI
Cloud Bigtable
BI
Looker